

Magnetic Hardening in Low-Dimensional Ferromagnets

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The key technical problem relevant to permanent magnets and magnetic recording media is how to enhance magnetic hardening for heavy-duty applications. Although long efforts have been made to understand coercivity mechanisms in the past century, many challenges remain in the fundamental and processing aspects in the research and development for stronger magnets. In the past two decades, we have pioneered in the area of nanoscale magnets and have made significant progress in understanding and processing novel nanostructured hard magnetic materials. We revealed that coercivity can be substantially enhanced in the low-dimensional structures via *anisotropy engineering* that leads to a solution to the outstanding Brown's Coercivity Paradox. It is also demonstrated that nanostructured high anisotropic magnets, including exchange-coupled nanocomposite magnets and nanowire-based magnets, have tremendous technical and economic advantages as the next generations of permanent magnets and magnetic recording media for future advanced applications.

Index Terms — Anisotropy, Coercivity, Permanent Magnets, Recording Media.

I. INTRODUCTION

HOW “hard” (coercive) a ferromagnet can be has been an important question for both fundamental research and applications in magnetic recording media and permanent magnets. Eight decades ago, William Fuller Brown [1] offered his famous theorem to correlate coercivity with the magnetocrystalline anisotropy fields in ferromagnetic materials. However, the experimental coercivity values have been far below the calculated levels given by the theorem, leading to so-called Brown's Coercivity Paradox [2] which remains unsolved even though researchers have made sustained efforts to understand it and to resolve it. Coercivity still cannot be predicted and calculated quantitatively by modelling.

II. SYNTHESIS OF HIGH K_1 NANOPARTICLES

Progress has been made in the past 20 years in understanding coercivity mechanisms in nanoscale low-dimensional ferromagnets synthesized via novel *bottom-up approaches*. Various ferromagnetic materials are investigated, including metals, alloys and ceramics. We systematically studied the geometric confinement in fine nanoparticles (0D), nanowires (1D), and thin layers (2D) that give rise to new magnetic effects, including the size, surface, and shape effects on magnetic properties, particularly on Curie temperature, magnetic anisotropy and coercivity.

As it is well known that ferromagnetism is a size-dependent physical phenomenon. However, nanoscale ferromagnetic samples with high magnetocrystalline anisotropy constant K_1 with controllable size and shape have not been available till recent times. By adopting newly invented methodologies including the salt-matrix annealing, surfactant-assisted milling [3], and improved hydrothermal and chemical solution techniques, we have successfully synthesized monodisperse ferromagnetic nanoparticles (FMNPs) including Fe–Pt, Fe–Co, and Sm–Co nanoparticles and Co nanowires with their

properties strongly size- and shape-dependent [3,4].

III. SIZE-DEPENDENT PROPERTIES

Monodisperse FMNPs with the desired size and shape, compositions, and structures have interesting magnetic properties and can be finely tuned for many different applications. We have studied anisotropy-dependent ferromagnetic properties of FMNPs, ranging from intrinsic magnetocrystalline anisotropy to extrinsic shape and surface anisotropy, and their effects on the magnetic hardening (coercivity). It has been quantified that the nanoparticle ferromagnetism is a size-sensitive and is effected substantially by geometric parameters of the material at nanometer scale.

We pioneered the synthesis and characterization of low-dimensional hard magnetic materials, including the bottom-up approaches to the production of nanocomposite magnets and the Co nanowires with extraordinarily high coercivity (Fig.1).[3,4] Chemical and physical production of hard magnetic nanoparticles enable size and shape control thus we are able to reduce the demagnetizing factors to minimum.

In 1945 Dr. W. Brown published his famous Brown's Theorem [1,2] that can be written in a simple formula:

$$H_c \geq H_a - N_d M_s \quad (1)$$

where H_c , H_a and M_s are the coercive field, the magnetocrystalline anisotropy field and the saturation magnetization, respectively. N_d is the demagnetization factor which is dependent on the magnet shape[2]. Since no experimental data could meet the theorem for seventy years, the discrepancy led to a term *Brown's Paradox*.

One of the major barriers to verify Brown's Theorem in experiments is the difficulty to meet the pre-condition (assumption) for the Theorem: the magnet must be homogeneous. Most conventional ferromagnetic materials are polycrystalline containing multiple domains which makes the magnets not strictly homogeneous. After the effort for more than two decades, we realized the extraordinary magnetic hardening in single-crystalline single-domain Co nanowires.

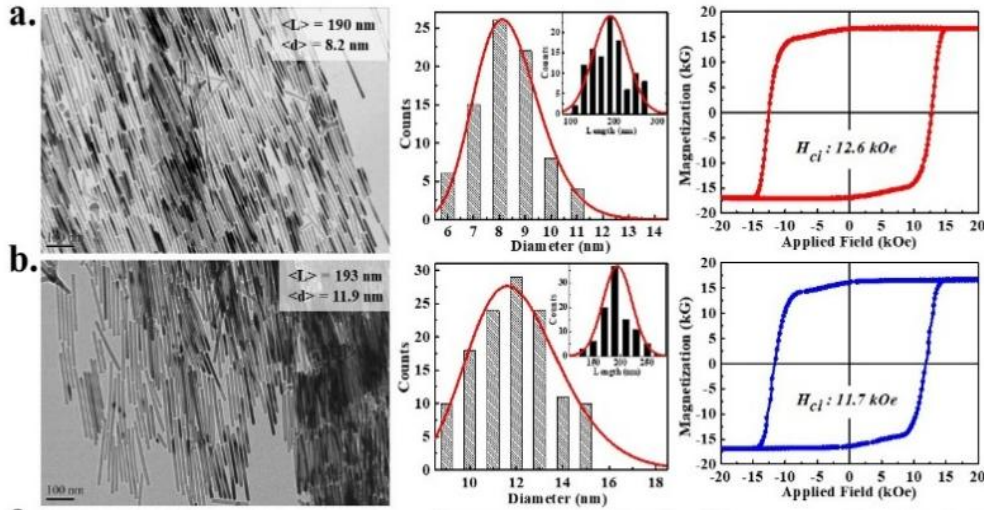


Figure 1 Diameter dependence of coercivity in Co NWs. (a, b) TEM images and the histograms of diameter and length distributions, and the room temperature hysteresis loops of Co NWs with diameters of ~ 8 and ~ 12 nm, with average length of ~ 190 nm. The obtained coercivity (H_{cl}) values are shown in the corresponding hysteresis plots.

Figure 1 shows the coercivity curves vs the nanowire diameter. The Co NWs with the diameter from 15 to 8 nm have the coercivity value for the aligned nanowires increases from 10.2 to 12.6 kOe. These coercivity values are substantially higher than the magnetocrystalline anisotropy of cobalt (7.6 kOe), which proves Brown's theorem experimentally for the first time.

IV. CONCLUDING REMARKS

If we define a *Brown magnet* as a piece of permanent magnets that conforms to Brown's Theorem (with its coercive field equal or higher than its $H_a - N_d M_S$ value), we can conclude that the Co nanowires with small diameters (Fig.1) are Brown magnets. It is significant to see that the quantitative correlation given by the Brown's Theorem between H_a and H_c can be verified in the nanoscale 1D magnets.

Our understanding of anisotropy and coercivity is far from the end. We recently discovered a new type of magnetic anisotropy, the *superstructure anisotropy* in magnetic nanoparticle assemblies[5]. Based on the progress in research in this area, we proposed to conduct *anisotropy engineering* to fabricate novel hard magnetic materials, including engineering of both magnetocrystalline anisotropy and shape anisotropy [6,7]. It is well known that commercial Alnico permanent magnets have demonstrated the importance and feasibility of shape anisotropy engineering. There is no doubt that by making use of both magnetocrystalline anisotropy and shape anisotropy, future permanent magnets and magnetic recording media with high energy product and high coercivity will be obtained using nanoscale low-dimensional hard ferromagnets as the building blocks.

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